

Feasibility Memo — Outline

Dataset: `yaleemmlc_admissionprediction_triage.csv`

Audience: Dr. De Fretias + ED Board (clinical, non-technical)

(a) One-sentence verdict

- Proceed with modelling, but only as a US-derived, adult-only baseline — not a Caribbean-ready deployment — given the missingness in the vitals and demographic fields, the unresolved race/ethnicity imbalance, and the single-site academic-hospital population.
- The stance is "proceed with caveats": the dataset is clean enough to build a first baseline on, but not yet validated for the deployment context.

(b) Dataset summary

- **Size:** 55,121 ED encounters × 225 columns after cleaning (the raw file has 226 columns before the export row-index is dropped).
- **Structure:** around 25 structured clinical/demographic/admin columns plus roughly 200 chief-complaint flags, one-hot and sparse —most cells are 0. The dataset is wide and sparse, not deep.
- **Population:** adults only (age ≥ 18, no paediatric cases), single-site US academic hospital, with US-specific `arrivalmode` and `insurance_status` categories.
- **Target:** `esi` (Emergency Severity Index), the `Triage_Level` — ordinal, 1 (most urgent) to 5 (least urgent), assigned at the door, with no missing labels.
- **Class balance:** dominated by ESI 2–3, with very few ESI 1 cases realistic for a hospital ED, but it means ESI 1 is the hardest class to model reliably.
- **Key column families:**
 - Vitals: heart rate, systolic and diastolic blood pressure, respiratory rate, oxygen saturation, temperature (recorded in Fahrenheit, not Celsius), glucose, and an oxygen-device flag distinguishing room air from supplemental oxygen.
 - Demographics: age, gender, ethnicity, race, language, religion, marital status, employment status, insurance status several of these are fairness-sensitive.
 - Leakage columns, excluded from modelling: disposition and previous disposition. Both are decided hours after triage, once workup is underway, so including them would let the future leak into a triage-time prediction.

(c) Top 3 quality concerns

1. Physiologically implausible vitals need correction rather than deletion. The cleaning pipeline flags out-of-range values for example, a heart rate outside 20–250 bpm or a systolic BP outside 50–300 mmHg as errors, sets them to missing, and then imputes with the median.
2. The scale and pattern of missingness varies by column family. Vitals, demographics, and the chief-complaint flags each carry different missingness profiles, and the imputation strategy (median for vitals, "Unknown" for blank categories, 0 for unrecorded complaint flags) is stated plainly rather than assumed.
3. There is a demographic and geographic mismatch with the intended deployment context. The sample is US, adult-only, and single-site, and the race and ethnicity distribution is imbalanced. If one group dominates the sample, the model may underperform for everyone else, and the sample may not resemble a Caribbean ED population.

(d) Top 3 reasons to proceed

1. The target is clean and fully populated. **esi** has no missing labels and sits entirely on a well-defined 1–5 ordinal scale, giving a real, learnable target to build toward.
2. The vitals plausibly separate acuity. Respiratory rate in particular is understood clinically as the earliest deterioration signal, and comparing vitals across ESI levels is expected to show meaningful separation between classes.
3. The chief complaints collapse into a manageable, clinically sensible structure. Despite roughly 200 raw flags, they group into around 12 body systems, and complaint volume concentrates in a few systems trauma and musculoskeletal, cardiovascular, and respiratory rather than scattering uniformly. This is tractable feature engineering, not an unworkable mess.

(e) Caveats

- Disposition and previous disposition are never used as model inputs. Both are outcomes decided after triage, and using them is leakage rather than signal.
- Temperature is recorded in Fahrenheit. Any downstream cleaning, thresholding, or clinician-facing chart states this explicitly, since a normal reading would otherwise read as hypothermia.
- Race, ethnicity, and insurance status (a proxy for access) are the columns most likely to encode inequity into a triage model and are monitored throughout the project. The ESI distribution within each race category is treated as a standing check rather than a one-off plot.
- Generalizability is the largest open question. The dataset is adult-only, single-site, and built on US arrival-mode and insurance categories that do not map cleanly onto a Caribbean ED. It supports building and validating a baseline modelling approach, not validating clinical performance in the deployment setting.

- Imputation choices are defensible but not neutral. Median-imputing vitals and filling categorical blanks with "Unknown" are reasonable defaults, and the memo states this plainly rather than implying the data was complete to begin with.